Question #1 of 74

Question ID: 1675256

Oskar Dieter manages a fund with exposures to both German equity and fixed income. The fund has a market value of €450m, with 60% exposure to equity and 40% to fixed income. The equity portfolio has a beta of 1.4 relative to the DAX 30, and the fixed-income portfolio has a modified duration of 12.6.

Dieter wishes to adjust the asset allocation to 70% equity and 30% fixed income using the following futures contracts:

DAX 30 Futures		**Euro Bund Futures**	
Futures price	13,400	Futures price	€174.50
		Contract size	€100,000
		CTD price	€99.90
Multiplier	€25	Conversion factor	0.5723
		CTD modified duration	9.86

Taking which of the following positions is *most likely* to achieve Dieter's required asset allocation?

A) Long 188 Dax 30 Index futures, short 329 Euro Bund futures.

B) Long 329 Euro Bund futures, short 134 Dax 30 Index futures.

C) Long 134 Dax 30 Index futures, short 576 Euro Bund futures.

Question #2 of 74

Question ID: 1675249

Tom Corser is the manager of the $140,000,000 Intrepid Growth Fund. Corser's long-term view of the equity market is negative, and as a result, his portfolio is allocated defensively with a beta of 0.85. Despite his negative long-term outlook, Corser thinks the market is temporarily mispriced and could rise significantly over the next few weeks. Corser has implemented tactical asset allocation measures in his fund sporadically over the years, and he thinks now is another time to do so. Because he likes his long-term holdings, he decides to use a futures overlay rather than trading assets to implement his view of the market. Corser decides he wants to increase the beta of his portfolio to 1.25. The appropriate futures contract has a beta of 1.03, and the total futures price is $310,000. What is the appropriate tactical allocation strategy for Corser to accomplish his objective?

A) Buy 373 equity futures contracts.

B) Sell 175 equity futures contracts.

C) Buy 175 equity futures contracts.

Question #3 of 74

Question ID: 1675252

A manager of $30 million in mid-cap equities would like to move half of the position to an exposure resembling small-cap equities. The beta of the mid-cap position is 1.0, and the average beta of small-cap stocks is 1.6. The betas of the corresponding mid- and small-cap futures contracts are 1.05 and 1.5, respectively. The mid- and small-cap futures prices are $260,000 and $222,222, respectively. What is the appropriate strategy?

A) Short 17 mid-cap futures and go long 17 small-cap futures.

B) Short 17 small-cap futures and go long 17 mid-cap futures.

C) Short 55 mid-cap futures and go long 72 small-cap futures.

Question #4 of 74

Question ID: 1675204

Matthew Ricky runs a charity and is expecting to receive a donation of $20,000,000 in six months' time. He wishes to temporarily hold the money in a deposit account for three months, earning $LIBOR + 50 basis points, while he decides how best to use the funds. Ricky is worried that interest rates may decline before the deposit is made and decides to hedge the risk using Eurodollar futures. The six-month Eurodollar future is quoted at 97.50, with a contract size of $1,000,000. Six months later, Ricky receives the donation and makes the deposit when three-month USD LIBOR is quoted at 2%. Ricky closes out his futures position at a price of 98. The return that Ricky earned on his deposit over the three-month period is *closest* to:

A) $125,000.

B) $75,000.

C) $150,000.

Question #5 of 74

Question ID: 1675247

Sheila manages a $100 million fixed-income portfolio. The portfolio duration is currently 4.9, and she would like to increase it to 5.7. She selects a swap with a net duration of 6.1. Based only on the information provided, which of the following statements is *least likely* correct?

A) Sheila should have a pay-floating/receive-fixed position in the swap.

B) Sheila should select a swap with a notional principal of approximately $13 million to achieve the desired duration.

C) Sheila should have a receive-floating/pay-fixed position in the swap.

Question #6 of 74

Question ID: 1675222

Shota Hideaki manages a Japanese equity fund with a market value of ¥21,712,000,000 and a beta relative to the Nikkei 225 of 0.75. Due to recent sales, a cash balance of ¥1,065,500,000 has built up, and Hideaki has become concerned about the cash drag. Hideaki wishes to create a cash overlay to equitize these funds. Hideaki is considering using Nikkei 225 futures and wishes the cash overlay to have the same beta as his existing portfolio. The current futures price is 21,500 with a multiplier of ¥1,000. To correctly execute the cash overlay, Hideaki would *most appropriately* purchase:

A) 44 contracts.

B) 50 contracts.

C) 37 contracts.

Question #7 of 74

Question ID: 1675255

Nicholas Avery manages a U.S. equity fund with a market value of $990 million. Currently, the fund is invested 70% in large-cap equity and 30% in small-cap equity, but Avery wishes to alter the allocation to 60% in large-cap and 40% in small-cap equity. Avery's large-cap portfolio has a beta of 1.2 relative to the S&P 500, and his small-cap portfolio has a beta of 0.9 relative to the MSCI USA Small Cap Index. Avery plans to make use of the following futures contracts to achieve his target asset allocation:

	S&P 500 Futures	MSCI USA Small Cap Index Futures
Futures price	3,100	720
Index multiplier	$250	$100

Taking which of the following positions is *most likely* to achieve Avery's target asset allocation?

A) Buy 1,650 MSCI USA small-cap futures; sell 115 S&P 500 futures.

B) Buy 1,375 MSCI USA small-cap futures; sell 128 S&P 500 futures.

C) Buy 1,238 MSCI USA small-cap futures; sell 153 S&P 500 futures.

Question #8 of 74

Question ID: 1675251

A manager of $40 million of mid-cap equities would like to move $5 million of the position to large-cap equities. The beta of the mid-cap position is 1.1, and the average beta of large-cap stocks is 0.9. The betas of the corresponding mid- and large-cap futures contracts are 1.1 and 0.95, respectively. The mid- and large-cap futures prices are $252,000 and $98,222, respectively. What is the appropriate strategy?

A) Short 29 mid-cap futures and go long 29 large-cap futures.

B) Short 20 mid-cap futures and go long 48 large-cap futures.

C) Short 23 mid-cap futures and go long 42 large-cap futures.

Question #9 of 74

Question ID: 1675211

If a manager shorts a forward currency contract to hedge the expected value of a foreign equity portfolio in one year, the worst-case scenario is if the portfolio's return is:

A) less than the expected value and the currency depreciates.

B) less than the expected value and the currency appreciates.

C) greater than the expected value and the currency appreciates.

Question #10 of 74

Question ID: 1675246

A manager of a $40 million fixed-income portfolio with a duration of 4.2 wants to lower the duration to 3. The manager chooses a swap with a net duration of 2.1. What notional principal (NP) should the manager choose for the swap to achieve the target duration?

A) $56,000,000.

B) $22,857,143.

C) $70,000,000.

Question #11 of 74

Question ID: 1675213

CorePro Incorporated, based in the United States, wishes to borrow €30m to finance a subsidiary in Italy. CorePro has decided to borrow domestically in dollars and then enter into a cross-currency basis swap to exchange the USD for euros. The basis on the euro-dollar swap is quoted as −20 basis points. The current $/€ exchange rate is $1.2. The swap has a 10-year tenor with quarterly settlement dates. At the last settlement date, 90-day USD LIBOR was 2%, and 90-day Euribor was 2.5%. The interest paid and received by CorePro at the next settlement date will be *closest* to:

A) Pays $180,000 and receives €172,500.

B) Pays €187,500 and receives $198,000.

C) Pays €172,500 and receives $180,000.

Question #12 of 74

Question ID: 1675205

Which of the following comments relating to fixed-income (bond) futures is *least accurate*?

A) The seller (short) party has the option of which bond and when to deliver in the delivery month.

B) The cheapest-to-deliver bond will have the highest basis and lowest repo rate.

C) The futures price is calculated using the cheapest-to-deliver bond and a conversion factor.

Gabriella Bright manages the pension fund of CCF Corporation. The fund maintains a portfolio with a 70% exposure to equity and 30% to bonds. Bright is allowed to temporarily change the exposure of the fund to take advantage of market dynamics.

The fund's equity portfolio has a beta of 1.025 and is equally invested in French and German stocks with betas of 1.20 and 0.85, respectively. Due to a recent popular uproar resulting from an increase in corporate scandals, the German government is considering implementing several measures that would effectively impose restrictions on investments. Bright expects that the German government will proceed with its plans, which would result in a capital outflow from the German market. Accordingly, using futures contracts, Bright wishes to decrease the exposure to German stocks by 10% and increase the exposure to French stocks by 10%. **Exhibit 1: Selected Information Pertaining to the Equity Index Futures Contracts (In EUR)** contains information about the available equity futures contracts.

Exhibit 1: Selected Information Pertaining to the Equity Index Futures Contracts (In EUR)

Current exposure of the equity portfolio to the German DAX	1.225 billion
Current exposure of the equity portfolio to the French CAC 40	1.225 billion
DAX futures price	12,500
DAX futures multiplier	25
CAC 40 futures price	6,500
CAC 40 futures multiplier	10

With regard to the pension fund's bond portfolio that holds German and French bonds, Bright expects German interest rates to decrease and French interest rates to increase. She decides to adjust the bond portfolio weights by using 290 German bond futures contracts

and 238 French bond futures contracts. One month later, the German bond yield decreases by 10 bps, while the French bond yield increases by 5 bps. **Exhibit 2: Selected Information Pertaining to the Bond Portfolio and the Futures Contracts** includes information pertaining to the bond portfolio and the futures contracts used.

Exhibit 2: Selected Information Pertaining to the Bond Portfolio and the Futures Contracts

Initial BPV of the French bonds	EUR 299,250
BPV of the cheapest-to-deliver (CTD) French bond	EUR 98.5
Conversion factor (CF) of the CTD French bond	0.97

Expecting the volatility of the DAX to increase, Bright considers purchasing a variance swap. She notes, "A variance swap is equivalent to a portfolio consisting of a long position in a basket of options and a short position in the underlying asset. Thus, a variance swap has a payoff that is convex in volatility and it is sold at the price of an at-the-money option."

Bright purchases a one-year variance swap on the DAX index with an initial volatility and strike price of 18% and a vega notional of EUR 45,000. Over the following year, the volatility of the DAX index increased by 2% ending up at 20%.

Question #13 - 16 of 74

Question ID: 1675264

Based on **Exhibit 1: Selected Information Pertaining to the Equity Index Futures Contracts (In EUR)**, to adjust the exposure of the pension fund's equity portfolio, Bright should *most likely*:

A) short 333 DAX futures contracts and long 2,262 CAC 40 futures contracts.

B) short 11,760 DAX futures contracts and long 16,019 CAC 40 futures contracts.

C) short 137 DAX futures contracts and long 660 CAC 40 futures contracts.

Question #14 - 16 of 74

Question ID: 1675265

The total change in the market value of the French bond exposure due to the increase in the French bond yield is *closest* to:

A) –€1,379,035.

B) –€1,496,250.

C) –€1,375,410.

Question #15 - 16 of 74

Question ID: 1675266

Bright's note pertaining to a variance swap is *most likely* incorrect with regard to:

A) the payoff of a variance swap.

B) the price of a variance swap.

C) the equivalence of a variance swap to a portfolio consisting of a long position in a basket of options and a short position in the underlying asset.

Question #16 - 16 of 74

Question ID: 1675267

The profit on the variance swap purchased by Bright on the DAX index is *closest* to:

A) €95,000.

B) €45,000.

C) €2,500.

Question #17 of 74

Question ID: 1675225

Which of the following statements regarding variance swaps is *least accurate*?

A) The payoff on a variance swap is determined by the difference between actual volatility squared over a given period versus implied volatility squared.

B) The payoff on a volatility swap is convex, resulting in gains that increase in relative magnitude as realized volatility increases.

C) The purchaser of a variance swap will profit if the realized volatility is greater than implied volatility.

Question #18 of 74

Question ID: 1675210

A maker of large computers has just received an order for some of its products. The agreed-upon price is in British pounds: £8 million. The firm will receive the pounds in 60 days. The current exchange rate is $1.32/£, and the 60-day forward rate is $1.35/£. If the firm uses the forward contract to hedge the corresponding exchange rate risk, how many dollars will it expect to receive?

A) $10,560,000.00.

B) $5,925,926.00.

C) $10,800,000.00.

Question #19 of 74

Question ID: 1675221

Sofia Chiara manages an equity fund that invests in Italian stocks. The fund has a market value of €120m at the start of the period. Chiara sold 563 futures contracts at the start of the period to give her overall portfolio a beta (relative to the FTSE MIB) of 0.8. The equity portion of her portfolio has a beta of 1.3. The FTSE MIB has a value of 21,350 at the start of the period, and the futures price was 21,315.

Twelve months later, the index has fallen to 21,050, and the futures price is 21,025. Assuming that the beta remains constant at 1.3 over the period and the futures contract multiplier is €5 per index point, the return on her fund at the end of the 12-month period is *closest* to:

A) −€1,347,537.

B) −€1,348,946.

C) −€1,375,687.

Question #20 of 74

Question ID: 1675217

A portfolio manager has a net long position in both stocks and bonds, and no cash. When attempting to replicate the existing portfolio, using bond and stock futures, which of the following statements is *most accurate*? The manager will:

A) go long on the stock futures, but short on the bond futures.

B) choose a single futures contract and net the bond and stock position.

C) go long on both stock and bond futures.

Question #21 of 74

Question ID: 1675200

Justin Nate wishes to speculate on the direction of interest rates, based on his belief that they will rise over the coming year. Specifically, Nate plans to use interest rate derivatives that are based on 30-day loan periods and that mature in 240 days. Current 240-day USD LIBOR is 5% and 270-day LIBOR is 5.2%. Which of the following statements is *least accurate*?

A) The futures price of the Eurodollar future is 99.45.

B) The forward price of the FRA and the forward rate implied by the Eurodollar future should be the same.

C) The current forward price for an FRA with a 30-day loan period that matures in 240 days is 6.6%.

Joanna Portobello is the director of a reputable wealth management boutique with a wide client base in the Middle East. The Middle East wealth management market has significant growth potential due to an increased appetite for moving assets from traditional bank accounts to more sophisticated financial securities.

Portobello is personally assigned to manage the portfolio of Tarek Karim, a VIP client. Karim's portfolio includes fixed-income and equity investments. Portobello advises Karim to use derivatives to temporarily amend his portfolio exposures, noting that the long-term market outlook is still unclear.

Karim's European fixed-income portfolio comprises semiannual corporate bonds that have the same maturity and coupon payment dates. He wishes to convert EUR to USD at each payment date and hedge against a potential depreciation of EUR. Portobello recommends using a currency swap.

The value of Karim's U.S. fixed-income portfolio is USD 100 million and the portfolio's current modified duration is 6. Through fixed-income futures contracts, Portobello targets a portfolio exposure of USD 150 million and a modified duration of 8. **Exhibit 1: Selected Information Pertaining to the Fixed-Income Futures Contracts** includes information pertaining to the related futures contracts.

Exhibit 1: Selected Information Pertaining to the Fixed-Income Futures Contracts

Modified duration of the cheapest-to-deliver (CTD) bond	9
CTD bond price	USD 130
Conversion factor (CF)	0.98
Futures settlement price	USD 100,000

Karim's U.S. equity portfolio is indexed to the S&P 500. Portobello mentions to Karim that she expects the U.S. equity market to be negatively affected by the recent changes in U.S. fiscal and monetary policies. To decrease the portfolio's exposure to the S&P 500 Index, Portobello suggests using an equity swap, wherein Karim will receive a floating rate and pay the return on the S&P 500 Index. **Exhibit 2: Selected Information Pertaining to the Equity Swap** includes information pertaining to the equity swap.

Exhibit 2: Selected Information Pertaining to the Equity Swap

Notional principal of the equity swap	USD 20 million
Expected annual floating rate that applies to the first settlement date	1.5%
Swap maturity	24 months
Swap settlement	Semiannual
S&P 500 value at the beginning of the swap	4,200
Expected S&P 500 value at the first settlement date	4,140

Due to political discord in Europe, Portobello expects an increase in market volatility. She suggests hedging against tail risk using a variance swap. She states, "A variance swap is a swap that involves net payments over its life that are based on the difference between the realized variance and the variance strike. Note that the closer the variance swap is to its maturity, the lower its sensitivity to changes in implied volatility."

Question #22 - 25 of 74

Question ID: 1675228

Which of the following statements regarding the currency swap is *least accurate*?

A) Karim is taking the euro coupon payments received from his bond investments and passing those through to the swap dealer in exchange for USD.

B) Currency risk is hedged in the swap.

C) There is an exchange of notional amounts at the beginning and ending of the swap.

Question #23 - 25 of 74

Question ID: 1675229

Based on Karim's current and target U.S. fixed-income exposure and **Exhibit 1: Selected Information Pertaining to the Fixed-Income Futures Contracts**, to increase the portfolio's exposure and modified duration, Portobello should *most likely* purchase:

A) 502 futures contracts.

B) 168 futures contracts.

C) 335 futures contracts.

Question #24 - 25 of 74

Question ID: 1675230

Based on **Exhibit 2: Selected Information Pertaining to the Equity Swap**, the expected net cash inflow on the first settlement date of the equity swap is *closest* to:

A) USD 0.59 million.

B) USD 0.14 million.

C) USD 0.44 million.

Question #25 - 25 of 74

Question ID: 1675231

Portobello's statement pertaining to a variance swap is *most likely*:

A) incorrect regarding the swap involving net payments over its life that are based on the difference between the realized variance and the variance strike.

B) incorrect regarding the swap's sensitivity to changes in implied volatility as the swap approaches its maturity.

C) correct.

Question #26 of 74

Question ID: 1675207

David Lizotte wishes to completely hedge his $40m U.S. fixed-income portfolio against parallel movements in the yield curve. David is considering using the following U.S. Treasury bond future:

U.S. Treasury Bond Future	
Futures price	$154.88
Futures contract size	$100,000
CTD	3.7%, 20 years to maturity
CTD price	$113.71
CTD conversion factor	0.7342
CTD modified duration	14.51

Lizotte's fixed-income portfolio currently has a modified duration of 12. The *most* appropriate futures contract transaction to fully hedge the portfolio would be to:

A) sell 291 contracts.

B) buy 291 contracts.

C) sell 214 contracts.

Question #27 of 74

Question ID: 1675198

Jeanne Leon is the portfolio manager for a U.S. fixed income fund. Leon is concerned that central bank policy will lead to rising interest rates over the next two years. In anticipation of this, Leon wishes to reduce the duration of his portfolio. Which of the following derivatives is *most likely* to achieve his goals?

A) Series of short FRA positions.

B) Two-year receiver swap.

C) Short position in Treasury bond futures.

Question #28 of 74

Question ID: 1675238

Michael Hallen, CFA, manages an equity portfolio with a current market value of $78 million and a beta of 0.95. Convinced the market is poised for a significant upward movement, Hallen would like to increase the beta of the portfolio by 40%, using S&P 500 futures currently trading at 856. The multiplier is 250. What is the number of futures contracts, rounded up to the nearest whole number, that will be needed to achieve Hallen's objective?

A) 144.

B) 143.

C) 139.

Question #29 of 74

Question ID: 1675240

Robert Zorn, CFA, manages an equity portfolio with a current market value of $150 million. The beta of the portfolio is 1.23, and Zorn is forecasting a short-term market adjustment that will significantly lower equity values and will occur in the near future. Zorn has decided to use S&P 500 futures, currently trading at 1,260, to reduce the portfolio's systematic risk exposure by 30%. The multiplier is 250. What is the number of futures contracts, rounded up to the nearest whole number, that will be needed to achieve Zorn's objective?

A) Sell 169.

B) Sell 176.

C) Buy 182.

Question #30 of 74

Question ID: 1675203

Ashley Catrina works for a U.K.-based private equity fund. A potential new portfolio company has been identified, and the fund is considering making the acquisition. Due to cash flow timings, the fund will require a 3-month £50m bridge loan commencing in nine months' time. Catrina wishes to lock in the interest rate on the bridge loan, so she sells 3-month Sterling futures maturing in nine months at 99.25. The contract size is £500,000. In nine months' time, Catrina initiates the loan at 0.90% and closes out the hedge at 99.10. The effective interest rate on the loan is *closest* to:

A) 0.90%.

B) 0.75%.

C) 0.80%.

Question #31 of 74

Question ID: 1675214

If the USD is trading at a positive basis, which of the following statements regarding cross-currency swaps is *least accurate*?

A) Additional returns can be earned on the USD relative to the interest rate suggested by covered interest rate parity.

B) A non-U.S. investor can earn returns greater than their domestic return by lending their domestic currency using a cross-currency swap and investing in the U.S. debt markets.

C) Covered interest parity must not hold.

Question #32 of 74

Question ID: 1675202

Eagle Jones purchased Eurodollar interest rate futures nine months ago. The contracts have now matured.

Contract size	$1,000,000
Nine-month futures price (at initiation)	96.45
Number of contracts purchased	10
Three-month USD MRR at futures' maturity	4.5%

Using the contract details, the profit on Jones's position at contract maturity is *closest* to:

A) −$9,500.

B) +$2,400.

C) −$23,750.

Question #33 of 74

Question ID: 1675197

A company has issued a 5-year semiannual-pay FRN with coupons equal to £LIBOR + 60 bp, with a par value of £50m. The company enters a 5-year semiannual interest rate swap, with notional principal of £50m, as the fixed-rate payer. The fixed rate in the swap is 4%. The 180-day £LIBOR at Day 360 is 5%. What is the net interest cost to the firm at the third settlement date on the swap?

A) £1,400,000.

B) £1,150,000.

C) £2,300,000.

Question #34 of 74

Question ID: 1675199

Andrew Jeremy is a U.K. fixed income manager who wishes to use interest rate swaps to increase the duration of his bond portfolio. The existing portfolio has a market value of £20m and a modified duration of 7. Jeremy wishes to increase the modified duration of his portfolio to 9 for the next two years. Two-year payer swaps currently have a duration of −5. What type of swap and what amount of notional principal will Jeremy require to achieve his stated goal?

A) Receiver swap notional principal £20m.

B) Receiver swap notional principal £8m.

C) Payer swap notional principal £20m.

Question #35 of 74

Question ID: 1675239

An investor has an $80 million diversified stock portfolio with a beta of 1.1. He would like to partially hedge his portfolio using S&P 500 futures contracts. The contracts are currently trading at $596.70 with a multiple of $250. Which of the following trades will reduce the portfolio beta by 50%?

A) Buy 295 contracts.

B) Sell 590 contracts.

C) Sell 295 contracts.

Question #36 of 74

Question ID: 1675250

Redden Capital Management manages an intermediate, high-quality bond portfolio with a value of $12 million and a modified duration of the portfolio is 4.4 years. Scott Stuart, the manager of the portfolio, is concerned about rising interest rates over the next few months and wants to make a tactical adjustment and cut the duration of the portfolio in half. Stuart asks Amy Swemba, a junior portfolio manager with Redden, to accomplish this task. Swemba is aware that a Treasury bond futures contract exists with a value of $102,000, with a modified duration of 8.2 years. Swemba replies to Stuart's comments with the following statements:

Statement 1:	The fastest and most cost-effective way to reduce the duration of the portfolio by half would be to sell $6 million dollars' worth of the actual bonds in the portfolio.
Statement 2:	The portfolio's duration could also be adjusted by selling 40 of the Treasury bond futures contracts.

After listening to Swemba's statements, Stuart should:

A) agree with Statement 1, but disagree with Statement 2.

B) disagree with Statement 1, but agree with Statement 2.

C) disagree with both Statement 1 and Statement 2.

Question #37 of 74

Question ID: 1675254

An investor has a $5 million investment in small-cap stocks. The investor enters into an equity index swap where the investor pays the return on the Russell 2000 and receives the return on the Dow Jones Industrial Average. The notional principal of the swap is $1 million. The resulting position is a synthetic mix of:

A) 20% large stocks and 80% small stocks.

B) 25% large stocks and 75% small stocks.

C) 16.67% large stocks and 83.33% small stocks.

Question #38 of 74

Question ID: 1675208

Samantha Holly is the fund manager of a German fixed-income fund. The fund has a market value of €90m and a modified duration of 14. Holly believes that the European central bank will end its policy of lower target interest rates and would like to reduce the duration of her portfolio. Holly has identified the following Bund future, which she would like to use:

Government Bond Futures	German Bund
Contract size	€100,000
Future price	€174.50
CTD price	€104.83
CTD conversion factor	0.5955
CTD modified duration	15.81

How many contracts will Holly need to sell to achieve a target duration of 8?

A) Sell 194 contracts.

B) Sell 326 contracts.

C) Sell 453 contracts.

Question #39 of 74

Question ID: 1675209

Sarah Finley wishes to hedge her portfolio of U.S. bonds against a rise in U.S. interest rates. Her portfolio has a weighted average maturity of 15 years. Sarah would *most appropriately* choose to:

A) enter into a receiver interest rate swap.

B) sell a strip of Eurodollar futures.

C) sell bond futures.

Question #40 of 74

Question ID: 1675223

Which of the following statements regarding the VIX Index is *most accurate*?

A) VIX is a measure of implied volatility derived from lookback options traded on the S&P 500.

B) VIX measures realized 3-month volatility on the S&P 500 Index.

C) VIX is the annualized standard deviation of the expected percentage changes in the index with 68% confidence.

Question #41 of 74

Question ID: 1675216

Which of the following statements about portfolio hedging is *least accurate*?

A) To synthetically create the risk/return profile of an underlying common equity security, buy the corresponding futures contract, sell the common short, and invest in a T-bill.

B) For a fixed portfolio insurance horizon, using put options generally requires less rebalancing and monitoring than with the use of futures contracts.

C) Futures contracts have a symmetrical payoff profile.

Question #42 of 74

Question ID: 1675253

An investor has a cash position currently invested in T-bills but would like to "equitize" it by using S&P futures contracts. Which of the following trades will create the desired synthetic equity position?

A) Selling S&P 500 futures contracts short.

B) Buying S&P 500 futures contracts.

C) Selling the T-bills and buying S&P 500 futures contracts.

Question #43 of 74

Question ID: 1675226

David Climo purchased a 12-month variance swap on the S&P 500 with a vega notional of $60,000 and a strike of 6%. Seven months have passed, and the S&P has a realized volatility of 6.5% over this period. The strike of a variance swap with five months to maturity is quoted at 5.8% at this time, and the 5-month USD LIBOR is 3%. The current value of the swap is *closest* to:

A) $31,250.

B) $13,300.

C) $13,136.

Jack Smith manages the endowment portfolio of IUL University. IUL's endowment portfolio includes both bond and equity investments. The university's board of directors decides to increase the amount of financial aid and number of scholarships offered to students. Accordingly, the spending rate from the endowment portfolio is expected to be much higher this year than last year. New donations are expected to be received, and Smith is planning to invest them in a bond portfolio that will provide regular cash inflows.

A few weeks after receiving the donations, which total $50 million, Smith uses the proceeds to buy floating-rate bonds. A few days later, the government announces that it is considering the implementation of a stimulus package, which would decrease market interest rates. Accordingly, to hedge against a potential decrease in interest rates, Smith implements an interest rate swap. Information pertaining to the bond portfolio and the interest rate swap is shown in **Exhibit 1: Selected Information Pertaining to the Floating-Rate Bond Portfolio and the Interest Rate Swap**.

Exhibit 1: Selected Information Pertaining to the Floating-Rate Bond Portfolio and the Interest Rate Swap

Portfolio value	USD 50 million
Current portfolio modified duration	0.4
Target portfolio modified duration	5.0
Interest rate swap modified duration	2.5

A few months later, the university enters into an agreement with an international nongovernmental organization (NGO) that promises to provide $500 million of grants in two months. The university plans to invest the funds for three months, after which the funds will be used to renovate its technology room. Smith expects the market rate to decrease further during the next two months, before receiving the grants, and he would like to now lock in the rate that will be received by the endowment. Accordingly, he buys 500 Eurodollar futures contracts trading at 97.6, each with a basis point value (BPV) of $25 and a face value of $1 million. Two months later, the fund invests the received grants at the 90-day LIBOR + 10 bps.

The agreement signed with the international NGO includes a clause specifying that another USD 600 million will be granted to the university under two conditions—namely, that the investment should be made in European bonds and should provide semiannual interest payments spread out over the investment period. The board of directors agrees to the condition and notifies Smith. Expecting the EUR to depreciate against the USD, Smith believes it will be good practice for a team of interns from the university to offer their suggested strategies to hedge against the expected depreciation of the EUR. The interns' suggestions are shown as follows:

Suggestion 1:	Enter a receive USD, pay EUR currency swap with semiannual settlements.
Suggestion 2:	Enter a receive EUR, pay USD currency swap with semiannual settlements.
Suggestion 3:	Sell a USD 600 million USD/EUR futures contract.

The university's spending rate turned out to be lower than projected. As a result, because the endowment has an excess cash balance, Smith considers increasing the endowment's portfolio exposure to the S&P 500 Index—which is expected to increase in value. He gathers the following information on an S&P 500 equity futures contract.

Exhibit 2: Selected Information Pertaining to the S&P Equity Futures Contract

Futures contract beta (β)	1

Futures contract price	3500
Futures contract multiplier	USD 250
Excess cash	USD 14.875 million

Question #44 - 47 of 74

Question ID: 1675259

Based on **Exhibit 1: Selected Information Pertaining to the Floating-Rate Bond Portfolio and the Interest Rate Swap**, the swap that Smith implemented to hedge against a potential decrease in interest rates is *most likely*:

A) a receiver swap with a notional principal of USD 92 million.

B) a payer swap with a notional principal of USD 100 million.

C) a receiver swap with a notional principal of USD 313 million.

Question #45 - 47 of 74

Question ID: 1675260

Upon receiving the $500 million in grants, if the 90-day LIBOR is 2%, the effective interest received on the grant portfolio is *closest* to:

A) USD 3,000,000.

B) USD 2,625,000.

C) USD 3,125,000.

Question #46 - 47 of 74

Question ID: 1675261

Which suggestion will *most likely* hedge the EUR-denominated investment funded by the $600 million in grants received from the international NGO?

A) Suggestion 2.

B) Suggestion 1.

C) Suggestion 3.

Question #47 - 47 of 74

Question ID: 1675262

Based on **Exhibit 2: Selected Information Pertaining to the S&P Equity Futures Contract**, to equitize the excess cash balance and increase the portfolio's exposure to the S&P 500, Smith should *most likely*:

A) short 17 futures contracts on the S&P 500 Index.

B) long 17 futures contracts on the S&P 500 Index.

C) long 4,250 futures contracts on the S&P 500 Index.

Question #48 of 74

Question ID: 1675220

Sofia Chiara manages an equity fund that invests in Italian stocks. The fund has a market value of €120,000,000 and a beta of 1.3 relative to the major Italian index (FTSE MIB). Chiara wishes to reduce the beta of her portfolio to 0.8 over the next 12 months. Chiara will use the FTSE MIB future with 24 months to maturity to achieve her goal. Contract details are as follows:

FTSE MIB Future	
24 months futures contract price	21,312
Multiplier	€5
Current index value	21,350

The futures position that Chiara will need to take to achieve her desired beta is *closest* to:

A) short 11 contracts.

B) short 2,815 contracts.

C) short 563 contracts.

Question #49 of 74

Question ID: 1675248

An investor has a $100 million stock portfolio with a beta of 1.2. He would like to alter his portfolio beta using S&P 500 futures contracts. The contracts are currently trading at 596.90. The futures contract has a multiple of 250. Which of the following is the correct trade required to double the portfolio beta?

A) Sell 804 contracts.

B) Buy 1,608 contracts.

C) Buy 804 contracts.

Georges Arniston manages an equity fund at VBC, an international investment bank. The fund is primarily invested in Australian and U.S. stocks. Arniston recently received the approval of the bank's investment committee to use derivatives when amending portfolio exposures or speculating on market movements.

The Australian stock portfolio includes growth and value stocks. The growth stock portfolio has a fair value of AUD 30 million and a beta of 0.95, while the value stock portfolio has a fair value of AUD 25 million and a beta of 1.20. Arniston would like to shift 40% of the value portfolio's assets to the growth portfolio using equity futures contracts. Information on the available futures contracts is shown in Exhibit 1.

Exhibit 1: Selected Information on Value and Growth Futures Contracts

Beta of the futures contract on value stocks	1.15
Beta of the futures contract on growth stocks	1.05
Price of the futures contract on value stocks	400
Price of the futures contract on growth stocks	500
Multiplier for both futures contracts	AUD 300

With regard to the U.S. stock portfolio, Arniston implements an overlay strategy to increase the beta of the portfolio (currently at 0.95), using five futures contracts priced at 3,500 at initiation and with a multiplier of USD 250. The portfolio currently has a market value of USD 15 million. The return on the U.S. stock market turns out to be 2% for the term of the futures contracts, and the settlement price of the future contracts is USD 3,570.

To combat the economic crisis, the U.S. government implements a financial stimulus package. The beneficiaries of the relief package are expected to use the proceeds to invest in the U.S. equity market. Arniston believes that the Federal Open Market Committee (FOMC) may increase the fed funds rate to prevent the stock market from overheating. He gathers the following information on a fed funds futures contract in Exhibit 2.

Exhibit 2: Selected Information on the Fed Funds Futures Contract

Current Fed funds target range	0.95%–1.1%
Price of the Fed funds futures contract	98.8
Possible hike of the Fed funds rate	30 bps

The chief investment officer at VBC notifies Arniston that the investment committee at the bank has reviewed the portfolio's performance for the last three months and is pleased with the profit achieved by the introduction of derivatives to the portfolio. The committee invites Arniston to the next meeting to discuss the use of derivatives related to volatility, such as futures contracts on VIX. In the meeting, Arniston notes, "Volatility futures contracts and swaps are particularly attractive for hedging the tail risk of an equity portfolio during the current expected downturn of the market. Because equity and volatility derivatives are negatively correlated, futures contracts should be bought to gain protection against the downturn in the equity market. Moreover, based on market prices, higher volatility is expected in the short term than in the long term."

Question #50 - 53 of 74

Question ID: 1675269

Based on Exhibit 1 and the information pertaining to the growth and the value portfolios, to temporarily shift 40% of the value portfolio's assets to the growth portfolio, Arniston should *most likely*:

A) short 26,087 futures contracts on value stocks and long 18,095 futures contracts on growth stocks.

B) short 87 futures contracts on value stocks and long 60 futures contracts on growth stocks.

C) short 18 futures contracts on value stocks and long 16 futures contracts on growth stocks.

Question #51 - 53 of 74

Question ID: 1675270

The effective beta of the U.S. stock portfolio after using the five futures contracts is *closest* to:

A) 1.01.

B) 0.66.

C) 1.24.

Question #52 - 53 of 74

Question ID: 1675271

Based on Exhibit 2, the probability of a 30 bp hike in the fed funds rate is *closest* to:

A) 0.66.

B) 0.33.

C) 0.58.

Question #53 - 53 of 74

Question ID: 1675272

Noting the market conditions mentioned in Arniston's note and assuming she buys futures contracts on VIX, her return on the futures contracts will *most likely*:

A) decrease given the negative roll yield.

B) increase given the positive roll yield.

C) not be affected by the roll yield.

Question #54 of 74

Question ID: 1675257

Alan Carney manages a fixed-income fund and is concerned that the Federal Open Market Committee will increase the target rate. Alan has collected the following data:

Current fed funds target	1.75%−2.00%
Fed funds futures price*	98

*Nearest futures contract maturity after the date of the next FOMC meeting

The implied probability of a 25 bp increase in the target rate at the next FOMC meeting is *closest* to:

A) 50%.

B) 12.5%.

C) 25%.

Question #55 of 74

Question ID: 1675244

A manager of a $20,000,000 portfolio wants to decrease beta from the current value of 0.9 to 0.5. The beta on the futures contract is 1.1 and the futures price is $105,000. Using futures contracts, what strategy would be appropriate?

A) Short 69 contracts.

B) Short 19 contracts.

C) Long 69 contracts.

Question #56 of 74

Question ID: 1675195

A U.S. firm that borrows dollars and uses a plain vanilla currency swap to obtain euros for an investment in Europe is *most likely* trying to:

A) lower borrowing costs.

B) increase the duration of the position.

C) create a synthetic pay-fixed dollar loan.

Question #57 of 74

Question ID: 1675215

If the value of a stock portfolio equals 16 times the futures price of the appropriate equity index contract and beta of the equity portfolio, and futures prices were equal, how many contracts would it take to reduce the beta of the equity index to zero?

A) A long position in 4 contracts.

B) A long position in 16 contracts.

C) A short position in 16 contracts.

Question #58 of 74

Question ID: 1675219

Marjorie Scarlett manages a U.K. equity portfolio with a market value of £20m and a beta relative to the FTSE 100 of 1.2. Scarlett wishes to increase her exposure to U.K. equities, so she takes the equity return receiver position in a 2-year equity index swap with a semiannual settlement. The swap is based on the FTSE 100, and it has a notional principal of £5m. The FTSE 100 at the start of the swap's life is 7,520, and 6-month £ LIBOR is 4%. At the time of the first settlement date, the FTSE 100 is at 7,600 and 6-month LIBOR is 4.2%. Assuming that Scarlett's existing portfolio's beta remains unchanged at 1.2, the £ return on the portfolio at the first settlement date will be *closest* to:

A) £230,510.

B) £208,510.

C) £255,319.

Question #59 of 74

Question ID: 1675196

A company has substantial bank loans and pays interest at LIBOR +80 bp. The central bank has signaled its intention to increase target interest rates over the next 12-month period. What type of interest derivatives would the company *most appropriately* enter into to mitigate its floating rate exposure?

A) A payer swap.

B) Long short-term interest rate futures.

C) Short FRA contracts.

Question #60 of 74

Question ID: 1675241

An S&P 500 Index manager knows that he will have $60,000,000 in funds available in three months. He is very bullish on the stock market and would like to hedge the cash inflow using S&P 500 futures contracts. The S&P 500 futures contract stands at 1,100.00, and one contract is worth 250 times the index. Which of the following is the *most accurate* hedge for this portfolio?

A) Sell 218 contracts.

B) Buy 284 contracts.

C) Buy 218 contracts.

Grid Co., a trading company headquartered in the United States, decides to expand its operations to Asia by opening Grid Asia, a new office in Singapore. Jim Xenjou, an experienced derivatives trader, joins Grid Asia. During the following year, Grid Asia generates a return beating the market, and Xenjou was appointed as Grid Asia's investment lead.

Last week, during a meeting between Xenjou and Grid Asia's investment committee, Xenjou's suggestion to short a 6-month futures contract on Singaporean Treasury bonds was approved. **Exhibit 1: Selected Information Pertaining to the Bonds That Can Be Delivered Under the Futures Contract** includes information on bonds that can be purchased to be delivered by Grid Asia under the terms of the futures contract.

Exhibit 1: Selected Information Pertaining to the Bonds That Can Be Delivered Under the Futures Contract

	Bond A	Bond B	Bond C
Cash dirty price at futures maturity	SGD 134	SGD 135	SGD 137
Futures* settlement price	SGD 138	SGD 140	SGD 142
Conversion factor (CF)	0.9910	0.9801	0.9756

*Size of futures contract is SGD 500,000.

Grid Asia's Singaporean corporate bond portfolio is valued at SGD 60 million. The portfolio has a modified duration of 5 and a basis point value (BPV) of SGD 30,000. In anticipation of higher interest rate levels, Xenjou receives the approval to short 22 Eurobond futures contracts; the cheapest-to-deliver (CTD) bond underlying the contracts has a modified duration of 8, a BPVCTD of SGD 90, and a conversion factor of 0.96. Two months later, the interest rate increases by 1.5%.

Xenjou would like to decrease the modified duration of Grid Asia's Japanese fixed-income portfolio from 7 to 4. He enters an interest rate swap upon receiving the approval of the investment committee. The market value of the Japanese fixed-income portfolio is JPY 5.81 billion. The swap is settled semiannually, will expire in 2 years, has a modified duration of –2, and has a fixed rate of 1.8%. Expecting the floating rate to be 1.85% at the first settlement date, Xenjou asks one of the associates to report the expected net payment on the first settlement date.

Doubting that the Japanese stock market is in a bubble, Xenjou purchases a Nikkei 225 VI futures contract—a volatility futures contract based on the volatility index (VI) of the Nikkei 225. He asks Olivia Goh, one of the associates, to approximate the roll yield on the contract. Goh finds it difficult to estimate the roll yield; instead, she reports that the difference between the Nikkei 225 VI futures price and the Nikkei 225 VI is positive.

Question #61 - 64 of 74

Question ID: 1675233

Based on **Exhibit 1: Selected Information Pertaining to the Bonds That Can Be Delivered Under the Futures Contract**, which of the three bonds should *most likely* be purchased to be delivered by Xenjou at the maturity of the futures contract?

A) Bond C.

B) Bond A.

C) Bond B.

Question #62 - 64 of 74

Question ID: 1675234

The decrease in the Singaporean fixed-income portfolio value that results from the 1.5% increase in the interest rate is *closest* to:

A) SGD 4.5 million.

B) SGD 7.2 million.

C) SGD 4.2 million.

Question #63 - 64 of 74

Question ID: 1675235

The expected net amount paid or received on the first settlement date of the interest rate swap is *closest* to:

A) JPY 4.4 million.

B) JPY 1.5 million.

C) JPY 2.2 million.

Question #64 - 64 of 74

Question ID: 1675236

The positive difference between the VIX front- month futures price and the VIX spot price *most likely* indicates that the market is in:

A) backwardation and the roll yield is positive.

B) contango and the roll yield is negative.

C) backwardation and the roll yield is negative.

Question #65 of 74

Question ID: 1675194

A firm has outstanding floating-rate debt on which they pay LIBOR + 200 basis points, and management expects interest rates to increase in the very near future. To create synthetic fixed-rate debt, the *best* strategy for the firm is to enter into a swap in which they:

A) pay fixed and receive floating.

B) pay floating and receive fixed.

C) receive floating and pay floating.

Question #66 of 74

Question ID: 1675206

Simon Feldberg sold treasury futures that have now reached their delivery date. Which of the three eligible bonds below should Feldberg deliver?

	Bond A	Bond B	Bond C
Bond coupon	5.7%	5.75%	5.8%
Bond maturity	20 years	19 years	18 years
Clean price	$144.18	$144.17	$143.96
Accrued interest*	$0	$0	$0
Conversion factor	0.9653	0.9719	0.9782
Futures settlement price		$148.75	
Contract size		$100,000	

Accrued interest* = accrued interest at futures delivery (assumed zero)

A) Bond B.

B) Bond A.

C) Bond C.

Question #67 of 74

Question ID: 1675237

A manager of a $10,000,000 portfolio wants to increase beta from the current value of 0.9 to 1.1. The beta on the futures contract is 1.2, and the futures price is $245,000. Using futures contracts, what strategy would be appropriate?

A) Short 7 contracts.

B) Long 7 contracts.

C) Long 11 contracts.

Question #68 of 74

Question ID: 1675218

Which of the following statements related to equity swaps is *least accurate*?

A) Equity swaps can be used to gain exposure to equity markets without incurring custody fees.

B) The fixed or floating rate cash outflow that the equity return receiver pays will always be offset by an inflow from the return on the equity index.

C) An investor gaining exposure to equity using swaps will not have voting rights in the underlying equity.

Question #69 of 74

Question ID: 1675243

To synthetically create the risk/return profile of an underlying common equity security:

A) buy the corresponding futures contract and borrow at the risk-free rate.

B) sell short the corresponding futures contract and invest in a T-bill.

C) buy the corresponding futures contract and invest in a T-bill.

Question #70 of 74

Question ID: 1675242

A portfolio manager knows that a $10 million inflow of cash will be received in a month. The portfolio under management is 70% invested in stock with an average beta of 0.8 and 30% invested in bonds with a duration of 5. The most appropriate stock index futures contract has a price of $233,450 and a beta of 1.1. The most appropriate bond index futures has a duration of 6 and a price of $99,500. How can the manager pre-invest the $10 million in the appropriate proportions?

A) Take a long position in 22 of the stock futures and 25 of the bond futures.

B) Take a short position in 25 of the bond futures and 22 of the stock futures.

C) Take a long position in 25 of the stock futures and 28 of the bond futures.

Question #71 of 74

Question ID: 1675245

A manager of a $2 million fixed-income portfolio with a duration of 3 wants to increase the duration to 4. The manager chooses a swap with a net duration of 2. The manager should become a:

A) receive-floating counterparty in the swap with a notional principal of $1 million.

B) pay-floating counterparty in the swap with a notional principal of $2 million.

C) pay-floating counterparty in the swap with a notional principal of $1 million.

Question #72 of 74

Question ID: 1675201

Kevin McMahon believes that Euribor will rise over the next six months and wishes to speculate on this view using three-month Euribor futures. McMahon enters into three futures contracts with 12 months to maturity that have the following characteristics:

Contract size	€1,000,000
12-month futures price	100.350

McMahon closes out his position after six months when the futures price is 99.95 for a contract with six months to expiry. The profit on McMahon's trade is *closest* to:

A) €3,000.

B) €2,250.

C) €1,500.

Question #73 of 74

Question ID: 1675212

Which of the following statements regarding cross-currency swaps is *least accurate*?

A) If the euro has negative basis against the dollar, the party paying euro interest will pay more than Euribor at each settlement date.

B) In a USD-EUR currency swap, the party receiving dollars at the swap's initiation will pay dollar interest over the life of the swap.

C) Principal payments are exchanged at the beginning of the swap's life, and again at the same exchange rate at the end.

Question #74 of 74

Question ID: 1675224

The VIX futures market is currently in backwardation. The VIX is currently at 40, front-month futures are priced at 35, and second-month futures are priced at 30. Assuming the term structure remains constant over the next two months, which of the following trades is *most likely* to result in a position that benefits from roll yield?

A) Buy the second-month future and sell the front-month future.

B) Buy the second-month future and sell the VIX spot.

C) Buy the front-month future and sell the second-month future.